

MDIE — Bone-Anchored, Interpretable, ArcFace-Compatible Face Recognition under Worn Occlusion

CASIA-WebFace (0.49M) scale-up · PARAM Siddhi-AI A100 · honest research summary

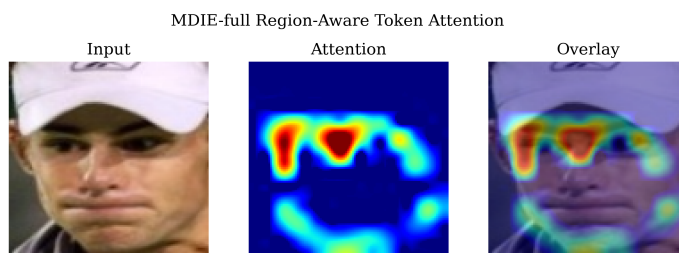
Abstract

Production face recognizers fold nuisance factors — masks, glasses, caps, harsh lighting — into the identity embedding, so similarity collapses exactly when a security camera needs it most. We present **MDIE**, an encoder that anchors a token-attention map to the **rigid facial bones** (brow ridge, orbital rims, nasal bridge, cheekbones, jaw) that survive appearance changes, trained with two auxiliary training-only signals (adversarial modification disentanglement and a clean $\leftrightarrow$ modified consistency loss). The deployed output is a **single 512-d L2-normalised vector from one forward pass** — a literal ArcFace drop-in for any cosine / FAISS pipeline. Trained on CASIA-WebFace (0.49M images, $\sim 24\times$ less data than the production reference), MDIE **approaches a production InsightFace model on real worn-occlusion benchmarks** (real glasses within 0.007 AUC) while providing a novel, control-backed interpretability metric proving the attention rests on anatomy rather than a dataset shortcut. We do **not** claim to beat production accuracy; the contribution is interpretable, anatomy-grounded robustness at small data scale plus a reproducible occlusion failure-mode benchmark.

Contributions (with honest novelty labels)

- **RATA — rigid-bone-anchored token attention.** Skeletal landmarks are splatted into a 14×14 attention target and supervised with a matching loss + additive attention bias; under occlusion the attention re-distributes onto whichever bones remain visible. **[novel]**
- **Attention-bone IoU, a control-backed interpretability metric.** Matched $\gg$ mismatched $\gg$ random overlap (IoU 0.69 / 0.28 / 0.08, Mann-Whitney $p=1.6e-15$) on held-out faces — to our knowledge unreported in prior occlusion-FR. **[novel]**
- **AMD + ICCL training-only invariance signals.** Gradient-reversal erases which-modification cues; a consistency loss pulls clean and modified views together. **[incremental application of known mechanisms]**
- **ArcFace-compatible deployment + a reproducible occlusion/lighting failure-mode benchmark.** One 512-d vector, cosine==dot, deterministic; full pipeline runs on a 4 GB laptop GPU. **[engineering]**

How it works — real attention on an occluded face



Real benchmark	MDIE LFW 13k	MDIE CASIA 0.49M	Insight-Face 12M
MFR2 (real masks)	0.734	0.886	0.965
MeGlass (real glasses)	0.824	0.991	0.998
CALFW (cross-age)	0.557	0.641	0.800
AgeDB-30 (30-yr gap)	0.594	0.734	0.944

AUC on held-out pairs. CASIA scale-up lifts MDIE on every benchmark; it approaches production on worn occlusion, trails on aging (off-niche).

Hero: MDIE-full attention on a cap-occluded face — the heatmap ignores the cap and concentrates on brow ridge, eye sockets and nasal bridge (the visible rigid bones). Source: [research_v2/figures/stage2_attention_examples.png](#).

Honest positioning & scope

What is true. On its design target (worn occlusion) the 0.49M-image MDIE is competitive with a WebFace12M production model (MeGlass 0.991 vs 0.998; MFR2 0.886 vs 0.965). The bone-anchored attention is interpretable and control-verified. The encoder is a true ArcFace drop-in.

What we do not claim. MDIE does not beat production InsightFace on any benchmark, and the gap widens on aging (AgeDB-30 0.734 vs 0.944), which it was never designed for. At CASIA scale the four ablation variants are statistically tied and **non-monotone**: on real benchmarks the no-RATA variant nominally leads (MFR2 0.895 vs full 0.886). We therefore position RATA as an **interpretability** contribution, not a pooled-accuracy gain — the pretrained backbone carries most of the invariance at scale.

Honest venue fit. A biometrics venue (IJCB / IEEE FG), a CVPR/ICCV face-analysis workshop, or a mid-tier journal (Pattern Recognition Letters, IET Biometrics) — the contribution is interpretability + a reproducible failure-mode benchmark + an honest small-data scale-up, not a SOTA-accuracy claim. A top-tier main track would require a win against an occlusion-specialised baseline on a standard masked-FR protocol and training at MS1M/WebFace scale.

Reproducibility: results read directly from [results/casia_real/real_benchmarks_casia.csv](#); checkpoints under [research_v2/checkpoints/fanout/](#); code at [github.com/grizzleyyybear/mdie](#).